



Inspiring Excellence

# Network Layer: IP Addressing

Lecture 7 | CSE421 – Computer Networks

Department of Computer Science and Engineering  
School of Data & Science

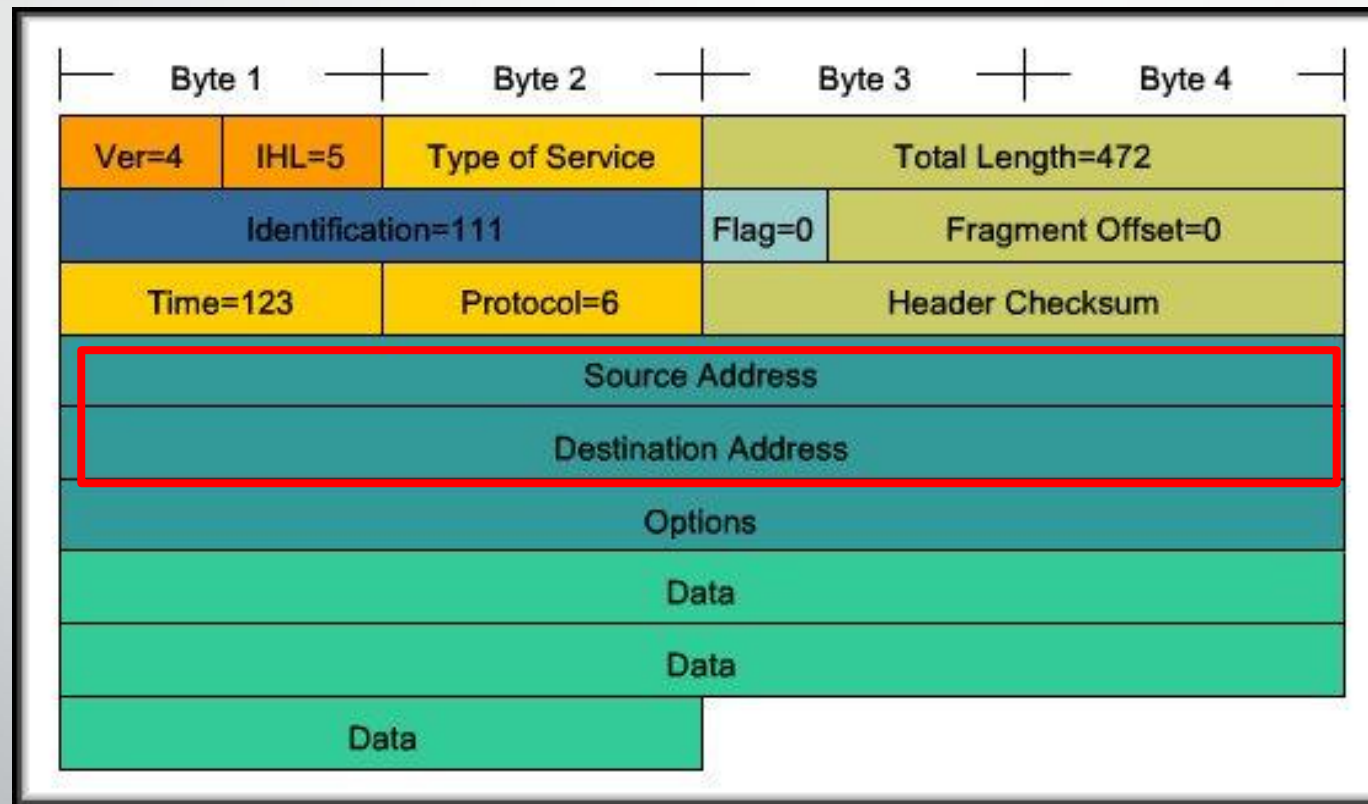
# Objectives

- IPv4 Address
  - Structure
  - Subnet/Prefix Mask
- Types of IPv4 Address
- IPv6 Address
  - Structure

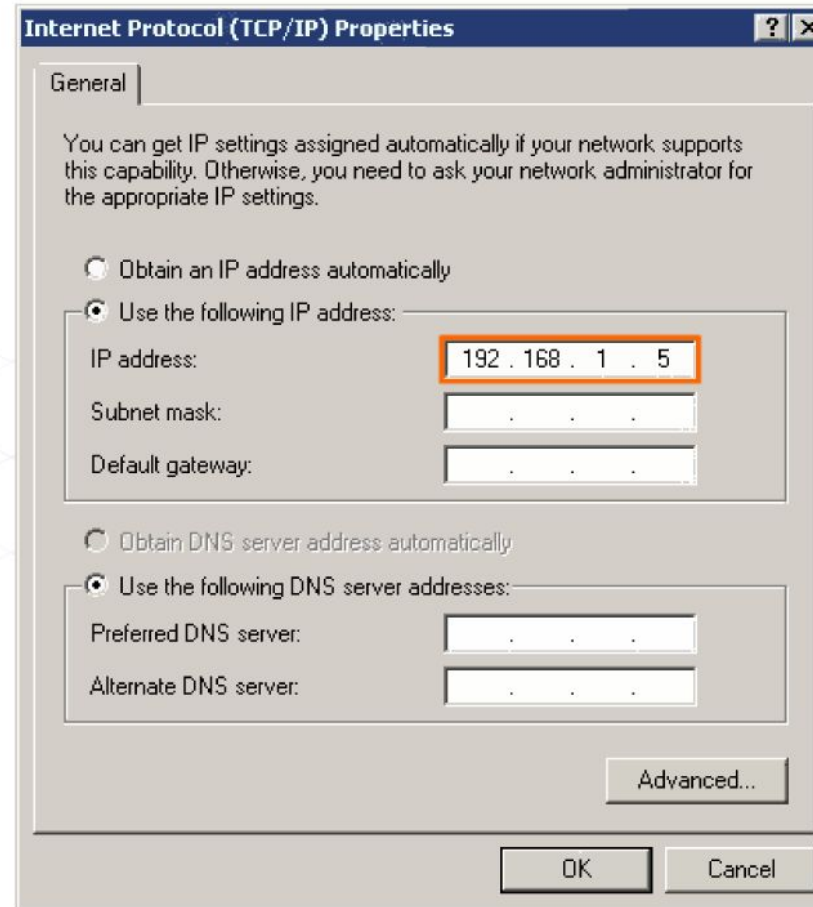
# Anatomy of IPv4

# Anatomy of an IPv4 Address

- Each device on a network must be uniquely identified at the Network layer.
- For IPv4, a 32 bit source and destination address is contained in each packet.



# IPv4 Addressing Structure



**Internet Protocol (TCP/IP) Properties**

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 5

Subnet mask: . . .

Default gateway: . . .

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

Advanced...

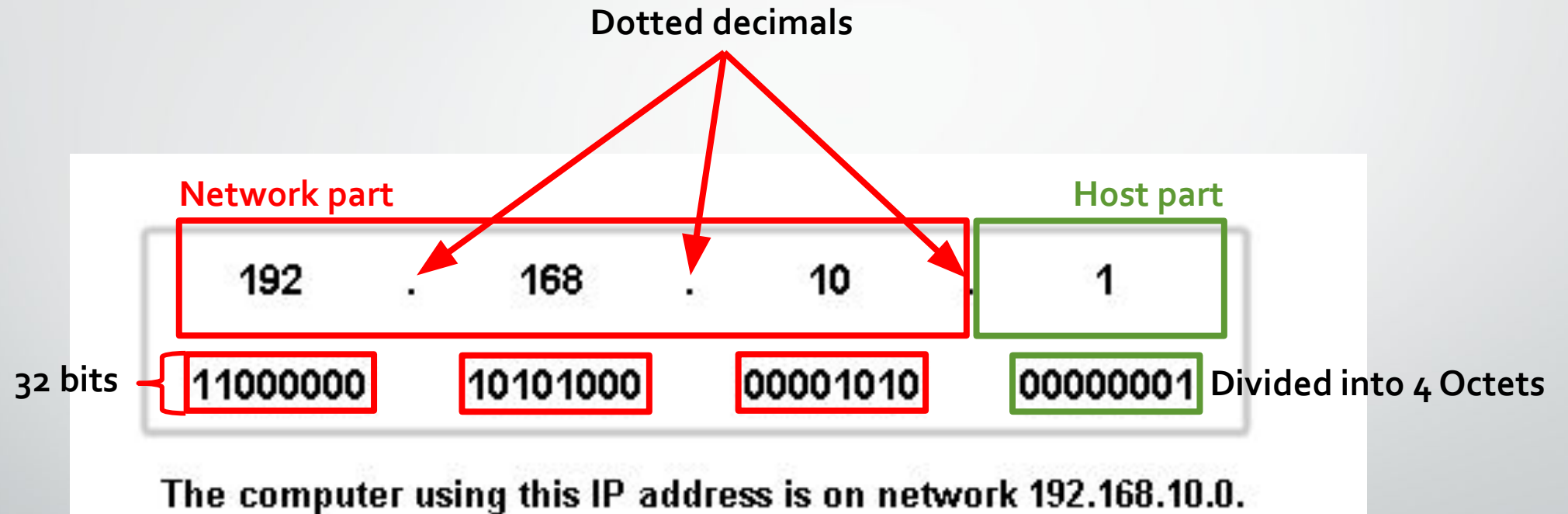
OK Cancel

I see you have  
assigned me  
an IP address  
11000000.1010  
1000.00000001.  
00000101  
Now other  
hosts can find  
me!



IP version 4 (IPv4) is the current form of addressing used on the Internet.

# Anatomy of an IPv4 Address



# Binary to decimal and Vice Versa

## Binary To Decimal Conversion

| Exponent                   | $2^7$                              | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|----------------------------|------------------------------------|-------|-------|-------|-------|-------|-------|-------|
| Position                   | 128                                | 64    | 32    | 16    | 8     | 4     | 2     | 1     |
| Bits                       | 1                                  | 1     | 1     | 1     | 0     | 1     | 0     | 1     |
| 1 BYTE / 1 Octet           |                                    |       |       |       |       |       |       |       |
| Add these numbers together | 128 + 64 + 32 + 16 + 0 + 4 + 0 + 1 |       |       |       |       |       |       |       |
| Decimal                    | 245                                |       |       |       |       |       |       |       |

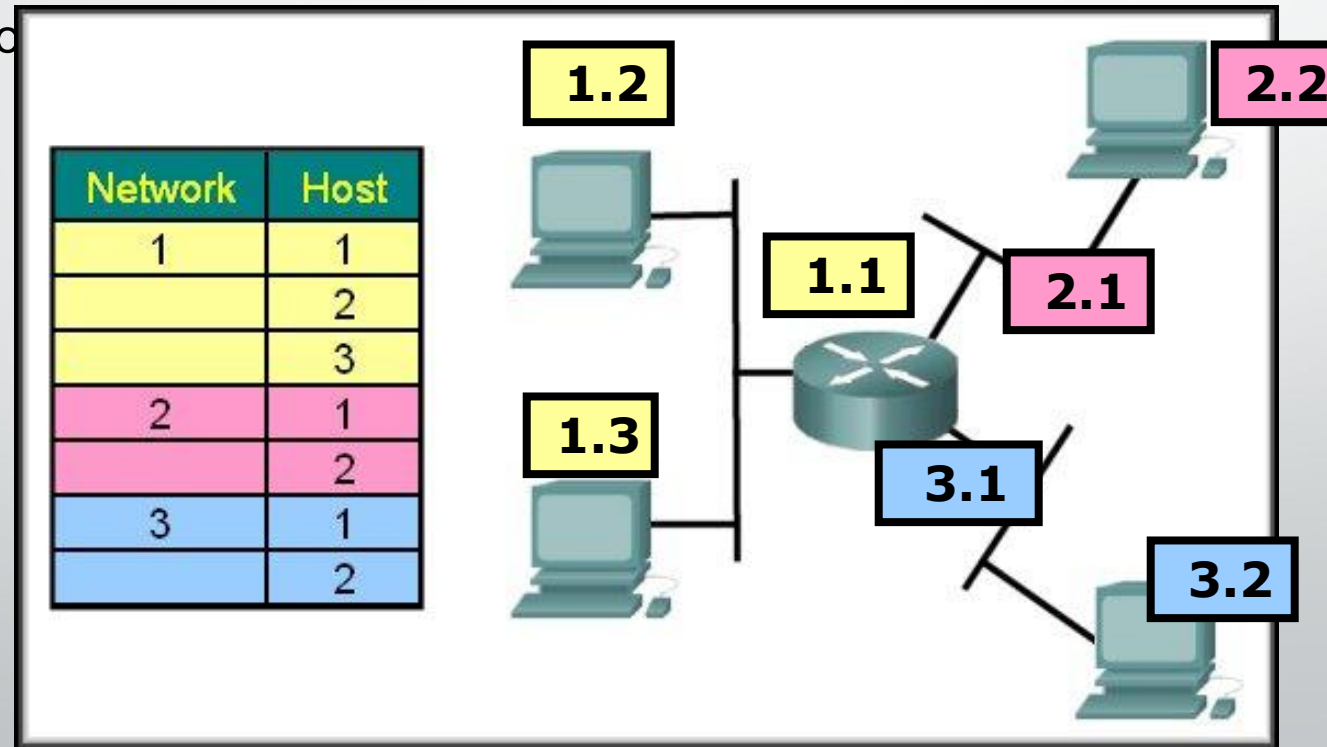
A 1 in this position means 64 is added to the total.

A 0 in any position means that 0 is added to the total.

11110101 in Binary = Decimal Number 245

# Networks and Hosts

- To identify a path or "route" through a network, the address must be composed of two parts:
  - Network** portion
  - Host** portion





# Network Portion

- Network Portion:**

- Some portion of the high-order bits
- A network can be defined as a group of hosts that have identical bit patterns in the network address portion of their addresses

| IP Address        | 192.     | 168.     | 1.       | 2        |
|-------------------|----------|----------|----------|----------|
| Binary IP Address | 11000000 | 10101000 | 00000001 | 00000010 |

|               |          |          |          |          |
|---------------|----------|----------|----------|----------|
| 192.168.1.2   | 11000000 | 10101000 | 00000001 | 00000010 |
| 192.168.1.67  | 11000000 | 10101000 | 00000001 | 01000011 |
| 192.168.1.204 | 11000000 | 10101000 | 00000001 | 11001100 |

# Network Portion

- **Host Portion:**

- A variable number of least significant bits that are called the **host portion** of the address.
- The **number of bits** used in this **host portion** determines the **number of hosts** that we can

| IP Address        | 192.     | 168.     | 1.       | 2        |
|-------------------|----------|----------|----------|----------|
| Binary IP Address | 11000000 | 10101000 | 00000001 | 00000010 |

|               |          |          |          |          |
|---------------|----------|----------|----------|----------|
| 192.168.1.2   | 11000000 | 10101000 | 00000001 | 00000010 |
| 192.168.1.67  | 11000000 | 10101000 | 00000001 | 01000011 |
| 192.168.1.204 | 11000000 | 10101000 | 00000001 | 11001100 |

# Prefix Mask

- How do we or devices identify the network part or the host part?

- Answer: Using the "Prefix Mask".

- 192.168.10.2/24

- Means that the first 24 bits are the network portion.
- The last 8 bits are the host portion.

- Subnet Mask is the other form of "Prefix Mask"

- Prefix length

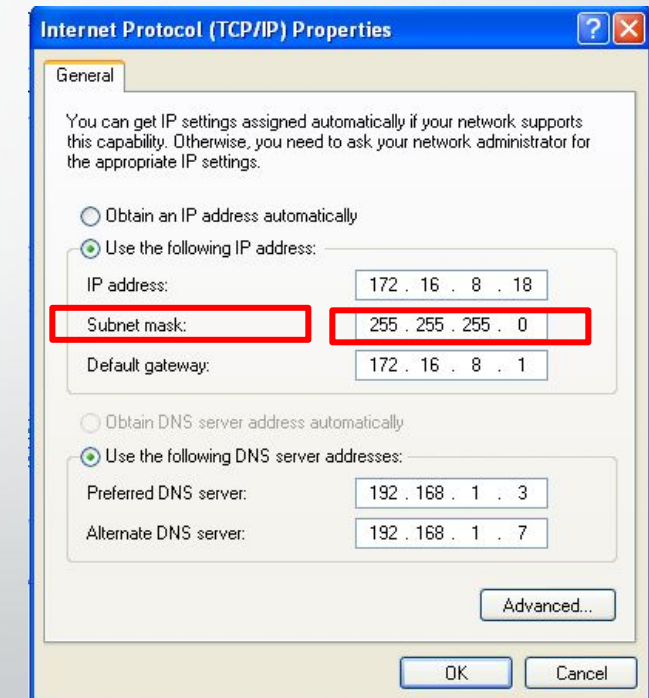
```
Z:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address                       . : 172.16.8.18
    Subnet Mask                       . : 255.255.255.0
    Default Gateway                   . : 172.16.8.1

Z:\>
```



# Subnet Mask

- The Prefix Mask and the Subnet Mask are different ways of representing the **same information**.
- **Conversion:**
  - Subnet mask has the **same format** as an IP address. Hence, it has **32 bits divided into 8 bits (octets)**
  - Prefix mask of **/24** means, **the first (MSB) 24 bits** of subnet mask would be **1** and the rest will be **0**
  - Binary: **11111111.11111111.11111111.00000000**  
 Decimal: **255.255.255.0**
  - Examples:
    - Prefix Mask of **/24** or a subnet mask of **255.255.255.0**
    - Prefix Mask of **/16** or a subnet mask of **255.255.0.0**
    - Prefix Mask of **/8** or a subnet mask of **255.0.0.0**

# Exercise

- What's the **subnet mask** of the following?
  - IP Address: 10.24.36.2 / 4
  - IP Address: 10.24.36.2 / 12
  - IP Address: 10.24.36.2 / 16
  - IP Address: 10.24.36.2 / 23
- What's the **prefix mask** of the following?
  - IP Address: 10.24.36.2; Subnet Mask: 255.255.224.0
  - IP Address: 10.24.36.2; Subnet Mask: 255.255.255.192
  - IP Address: 10.24.36.2; Subnet Mask: 255.255.255.252
  - IP Address: 10.24.36.2; Subnet Mask: 255.254.0.0

# ANDing the Binaries

- Inside data network devices, digital logic is applied for their interpretation of the addresses.
- AND is used in determining the network address.
  - $0 \text{ AND } 0 = 0$
  - $1 \text{ AND } 0 = 0$
  - $1 \text{ AND } 1 = 1$

|                 | Decimal     | Binary                              |
|-----------------|-------------|-------------------------------------|
| IP Address      | 135.15.2.1  | 10000111 00001111 00000010 00000001 |
| Subnet Mask     | 255.255.0.0 | 11111111 11111111 00000000 00000000 |
| Network Address | 135.15.0.0  |                                     |

# But Why AND?

- **Routers** use the **ANDing** process to determine the route a packet will take.
- The network number of the **destination IPv4 address** is used to find the network in the routing table.
- The router then determines the best path for the frame.

# IPv4 Addresses

**Network Address**

**Broadcast Address**

**Host Address**



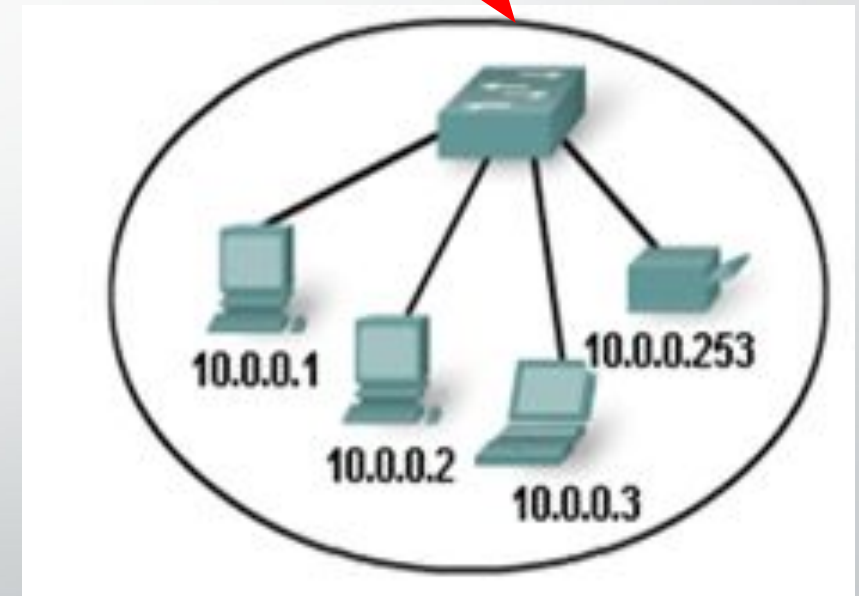
# The Addresses

## • Network Address

- All hosts in the network will have the same network bits.
- Cannot be assigned to a device.
- All host bits in this address will be **zero**.

| IP Address           | 10.      | 0.       | 0.       | 0         |
|----------------------|----------|----------|----------|-----------|
| IP Address in Binary | 00001010 | 00000000 | 00000000 | 00000000  |
| Network Part         |          |          |          | Host Part |

10.0.0.0/24



# The Addresses

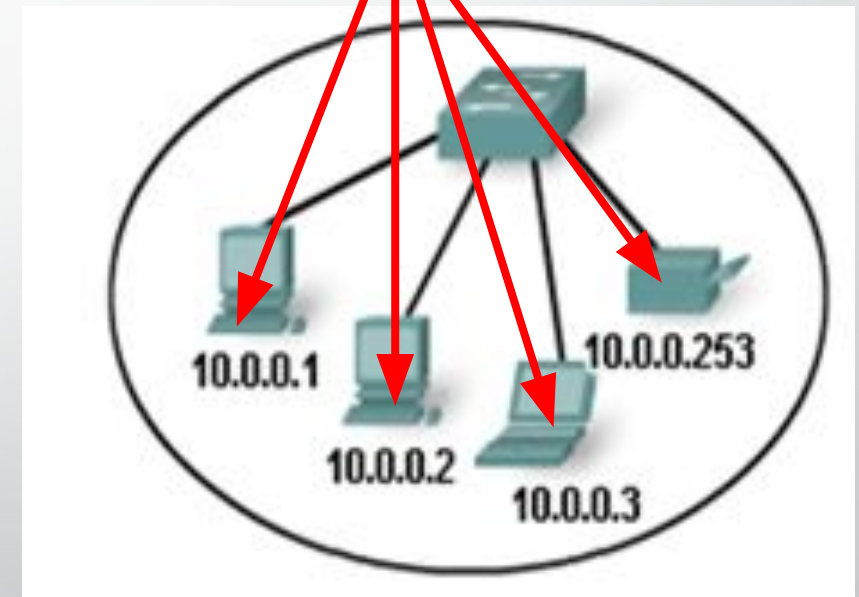
## Broadcast Address

- Used to send message to all hosts in the network using one single address
- Cannot be assigned to a device.
- All host bits in this address will be **one**.

10.0.0.255/24

|                      |              |          |          |           |
|----------------------|--------------|----------|----------|-----------|
| IP Address           | 10.          | 0.       | 0.       | 255       |
| IP Address in Binary | 00001010     | 00000000 | 00000000 | 11111111  |
|                      | Network Part |          |          | Host Part |

**Broadcast Address of 10.0.0.0/24 network is 10.0.0.255/24**



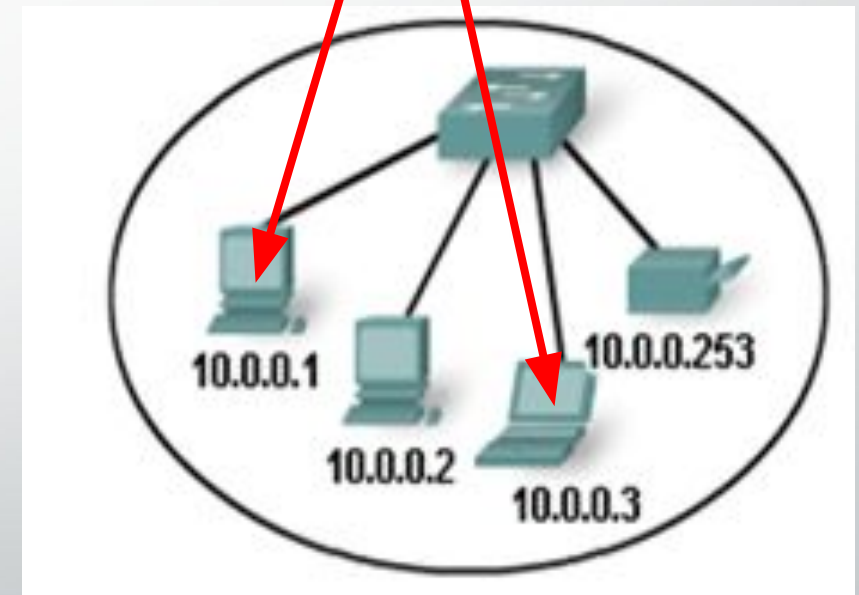
# The Addresses

## Host Address

- The unique address assigned to each device on the network.
- For a network of 10.0.0.0/24
  - Addresses 10.0.0.1 through 10.0.0.254 are
  - All host bits in this address will vary.

|                      |          |          |          |           |
|----------------------|----------|----------|----------|-----------|
| IP Address           | 10.      | 0.       | 0.       | 1         |
| IP Address in Binary | 00001010 | 00000000 | 00000000 | 00000001  |
| Network Part         |          |          |          | Host Part |
| IP Address           | 10.      | 0.       | 0.       | 3         |
| IP Address in Binary | 00001010 | 00000000 | 00000000 | 00000011  |

## Host Address



# The Addresses

- Say, you have a random IP address  
**192.168.10.193/24** or given as  
**192.168.10.193 255.255.255.0**
- Say, you have a random IP address  
**200.32.16.192/26** or given as  
**200.32.16.192 255.255.255.192**

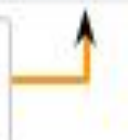
# Network Prefix

- The network prefix is not always /24.

Using Different Prefixes for the 172.16.4.0 Network

| Network        | Network address | Host range                | Broadcast address |
|----------------|-----------------|---------------------------|-------------------|
| 172.16.4.0 /24 | 172.16.4.0      | 172.16.4.1 - 172.16.4.254 | 172.16.4.255      |
| 172.16.4.0 /25 | 172.16.4.0      | 172.16.4.1 - 172.16.4.126 | 172.16.4.127      |
| 172.16.4.0 /26 | 172.16.4.0      | 172.16.4.1 - 172.16.4.62  | 172.16.4.63       |
| 172.16.4.0 /27 | 172.16.4.0      | 172.16.4.1 - 172.16.4.30  | 172.16.4.31       |

**SAME NETWORK ADDRESS  
ALL PREFIXES**

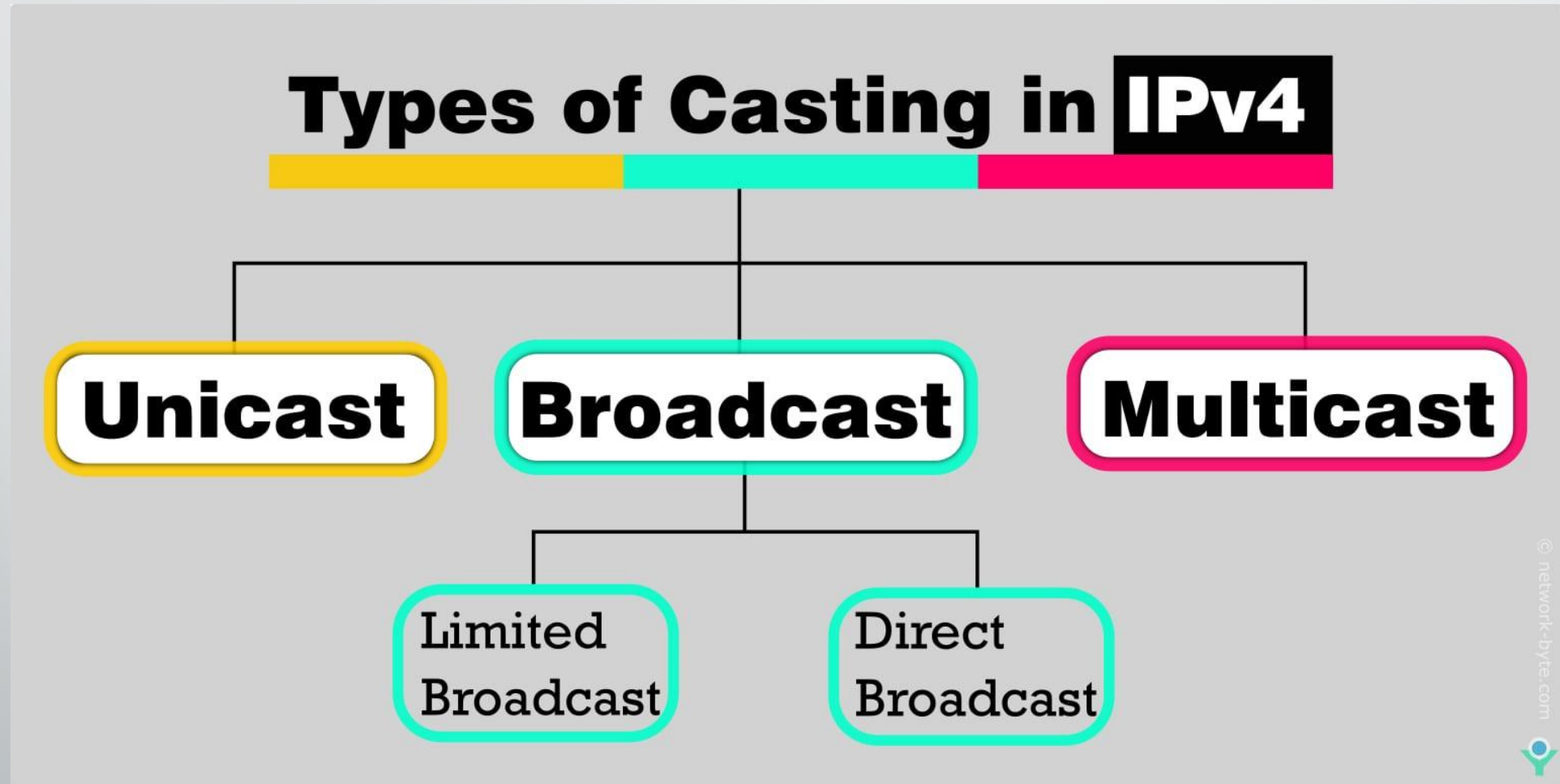


**DIFFERENT BROADCAST  
ADDRESS EACH PREFIX**



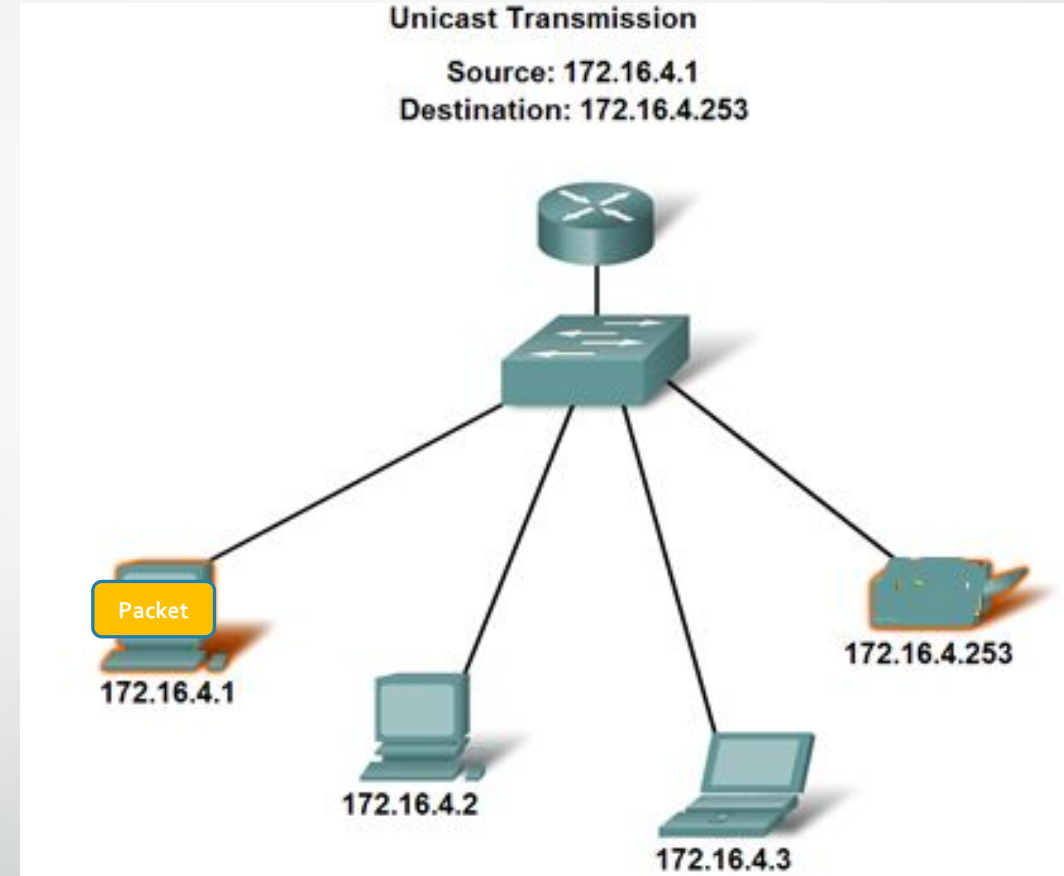
# Types of IPv4 Addresses

# Special Addresses



# Unicast

- **Unicast**
  - A message to one host.
  - Individual IPv4 addresses



172.16.4.253

172.16.4.1

Data

Dest IP Add

Source IP Add

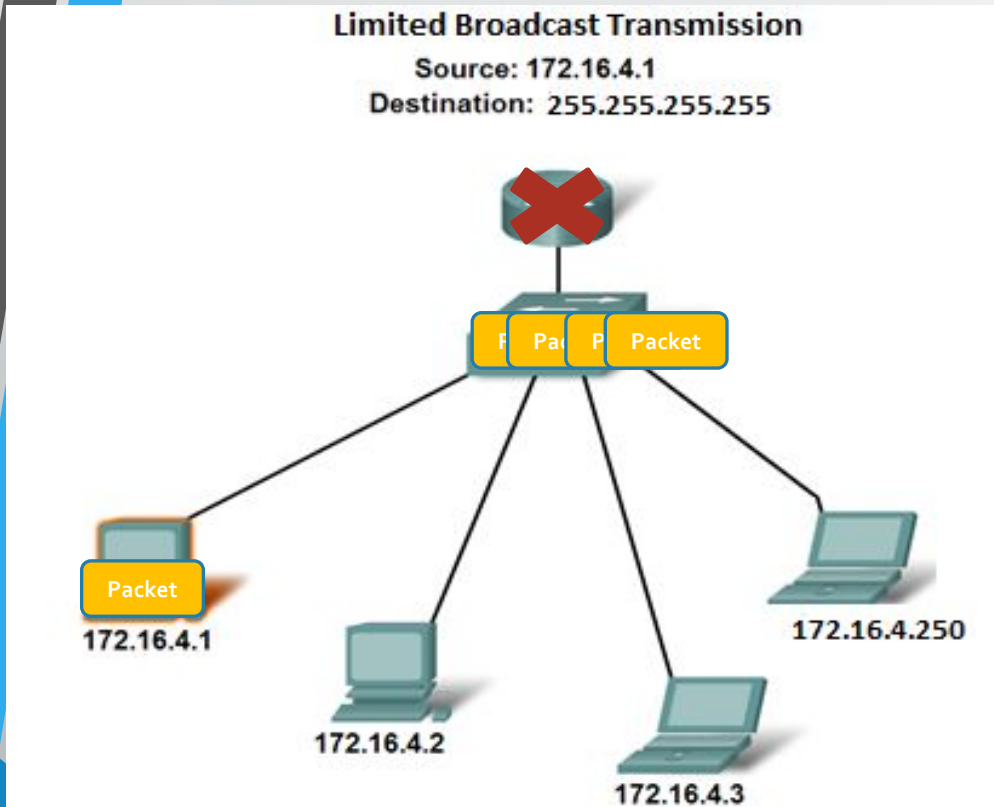
Packet



# Broadcast

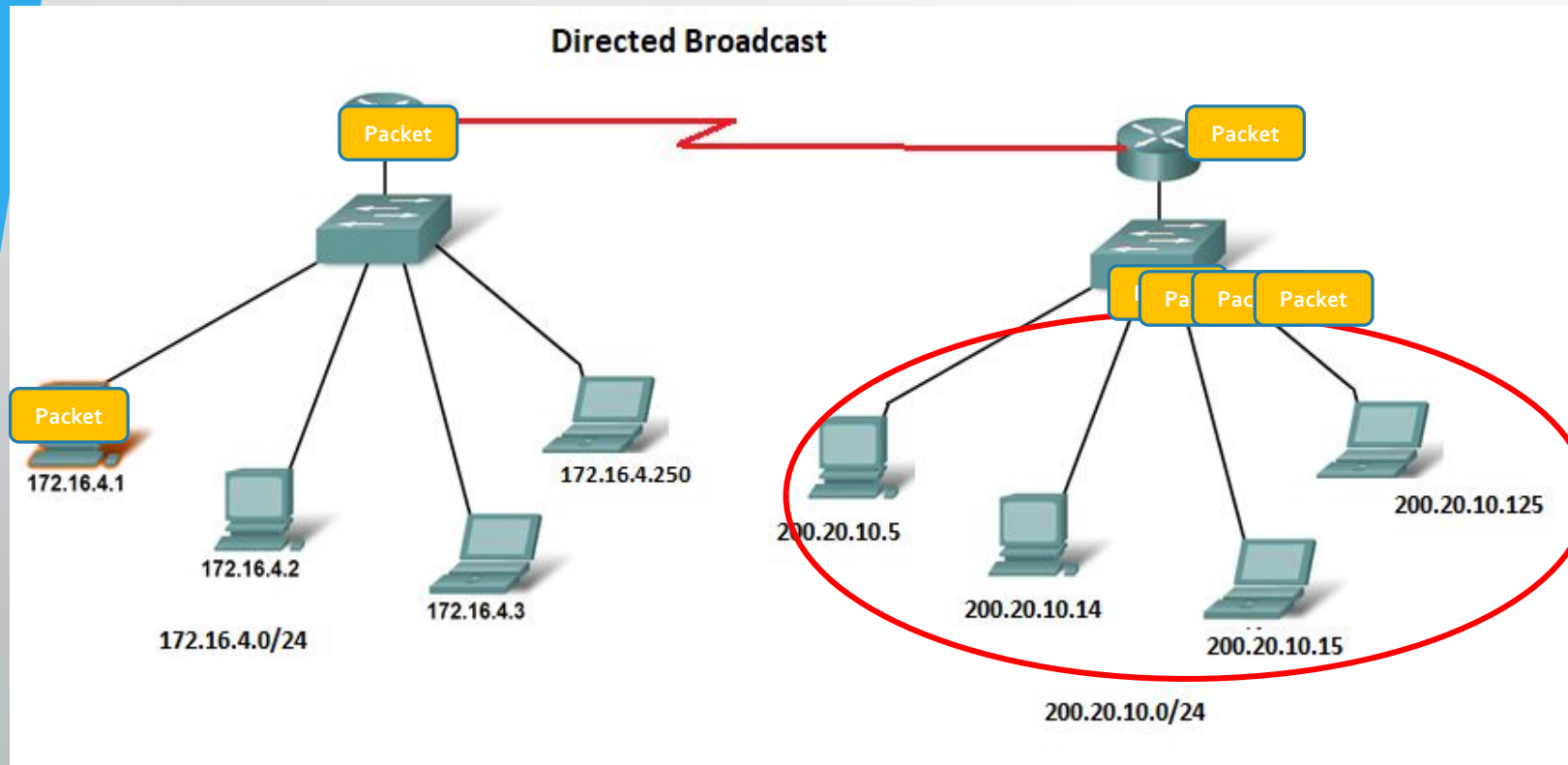
- **Limited Broadcast**

- A message to all hosts on the same physical/local network or subnet.
- **255.255.255.255**
- Never forwarded by routers!



| 255.255.255.255 | 172.16.4.1    | Data   |
|-----------------|---------------|--------|
| Dest IP Add     | Source IP Add | Packet |

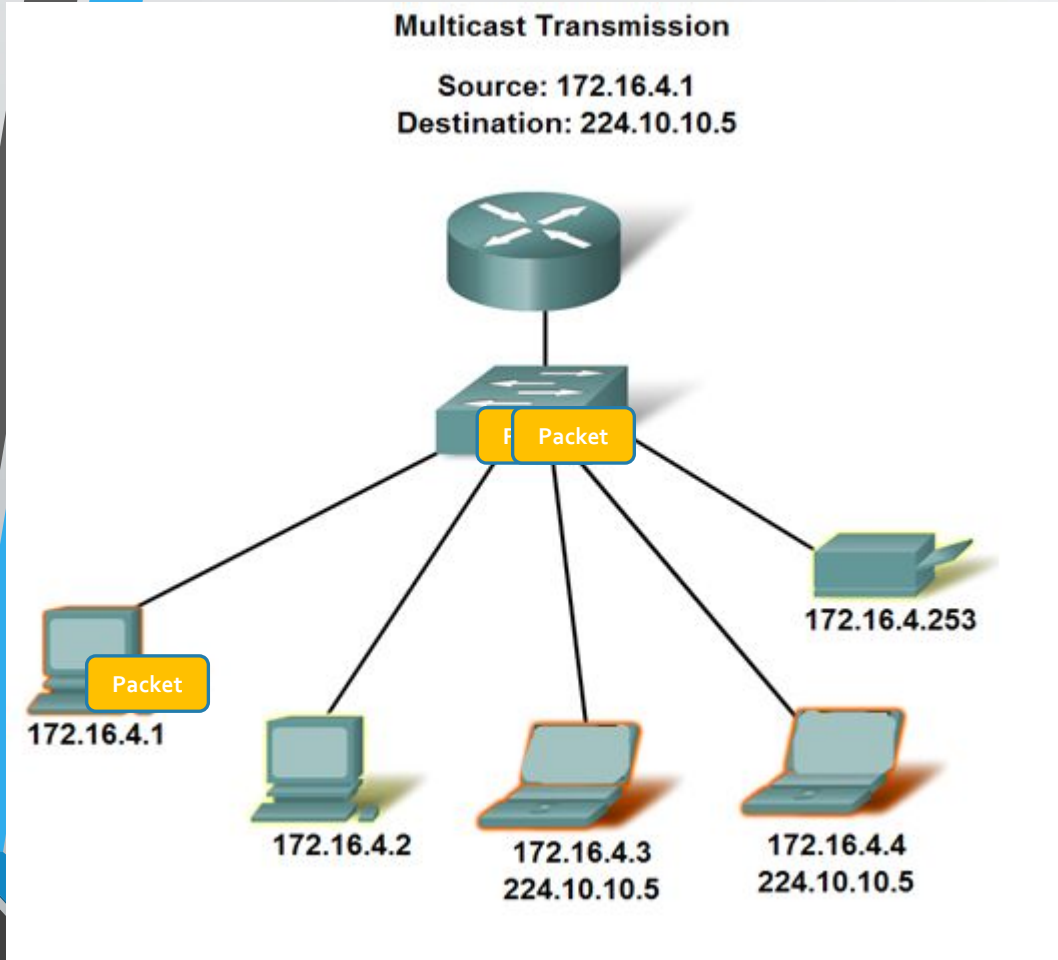
# Broadcast



|               |               |        |
|---------------|---------------|--------|
| 200.20.10.255 | 172.16.4.1    | Data   |
| Dest IP Add   | Source IP Add | Packet |

- **Directed Broadcast**
  - A message to all hosts on a different network or subnet.
  - **broadcast address of a network**
  - Example :  
200.20.10.255

# Multicast



## • Multicast Addresses

- A message addressed to a group of hosts.
- Uses an IP address starting within this range of **224 - 239**

## • Examples of Multicast Application

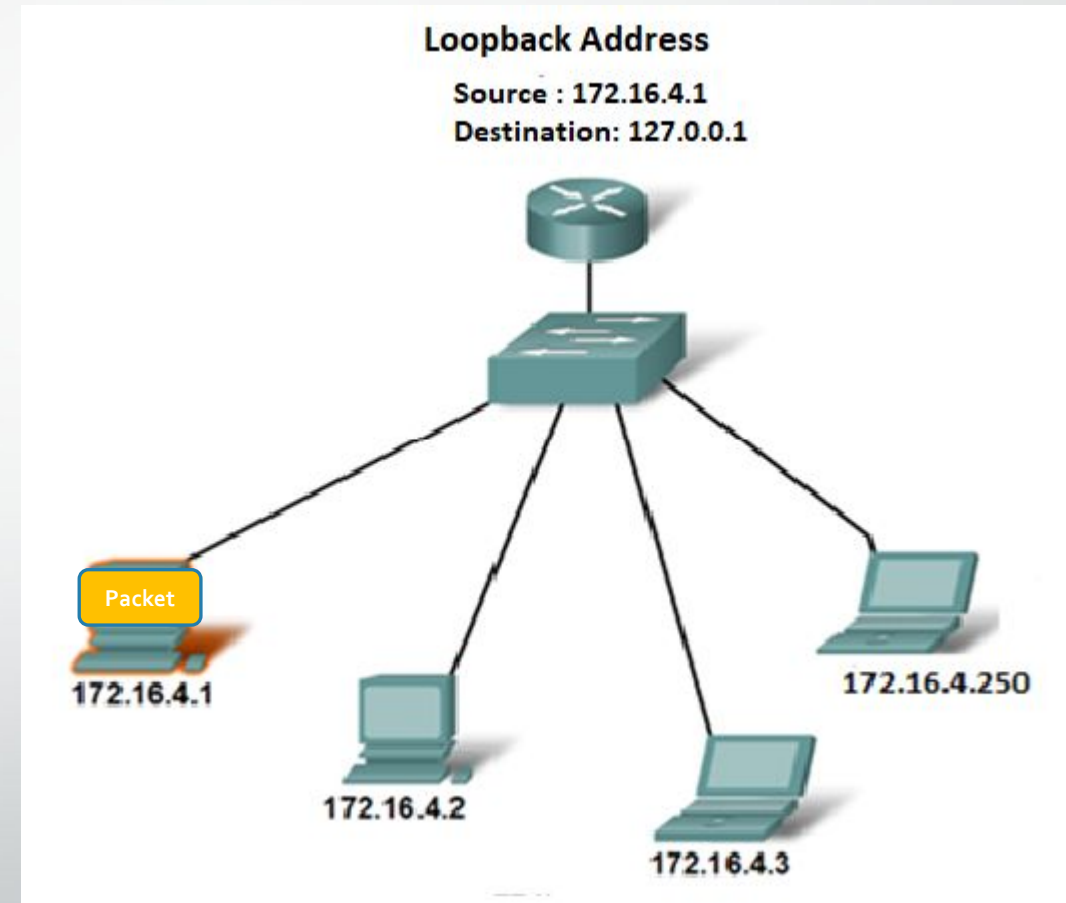
- Video and audio broadcasts
- Distribution of software
- News feeds

|             |               |        |
|-------------|---------------|--------|
| 224.10.10.5 | 172.16.4.1    | Data   |
| Dest IP Add | Source IP Add | Packet |

# Loopback

## • Loopback Address

- A message addressed to loop back in the device itself.
- **127.x.x.x** of **127.0.0.0/8**
- Not assigned to any device
- Testing and Troubleshooting purpose



|             |               |        |
|-------------|---------------|--------|
| 127.0.0.1   | 172.16.4.1    | Data   |
| Dest IP Add | Source IP Add | Packet |

# Anatomy of IPv6 Address

# Reasons for using IPv6

- Address Availability:

- IPv4: 32 bits - 4 octets

- $2^{32}$  or 4,294,467,295 IP Addresses.

- IPv6: 128 bits - 16 octets

- $2^{128}$

- $3.4 \times 10^{38}$  or

340,282,366,920,938,463,463,374,607,431,768,211,456

(340 undecillion) IP Addresses.



Every grain of sand on every beach on Earth could be assigned over a million unique IPv6 addresses, with plenty to spare (assuming approximately  $7.5 \times 10^{18}$  grains of sand globally and  $2^{128}$  IPv6 addresses).

# IPv6 Address

- 128 bits
- given below is a 128 bit IPv6 address represented in binary

```
0010000000000001 0000110110111000 1010110000010000 1111111000000001  
0000000000000000 0000000000000000 0000000000000000 0000000000000000
```

- Each 4 bits is converted into a Hexadecimal digit
- Each block contains 4 Hexadecimal digits

Each block is separated by ':' symbol



# IPv6 Address

0010000000000001 0000110110111000 1010110000010000 1111111000000001  
0000000000000000 0000000000000000 0000000000000000 0000000000000000

2 0 0 1

2001 : 0db8 : ac10 : fe01 : 0000 : 0000 : 0000 : 0000

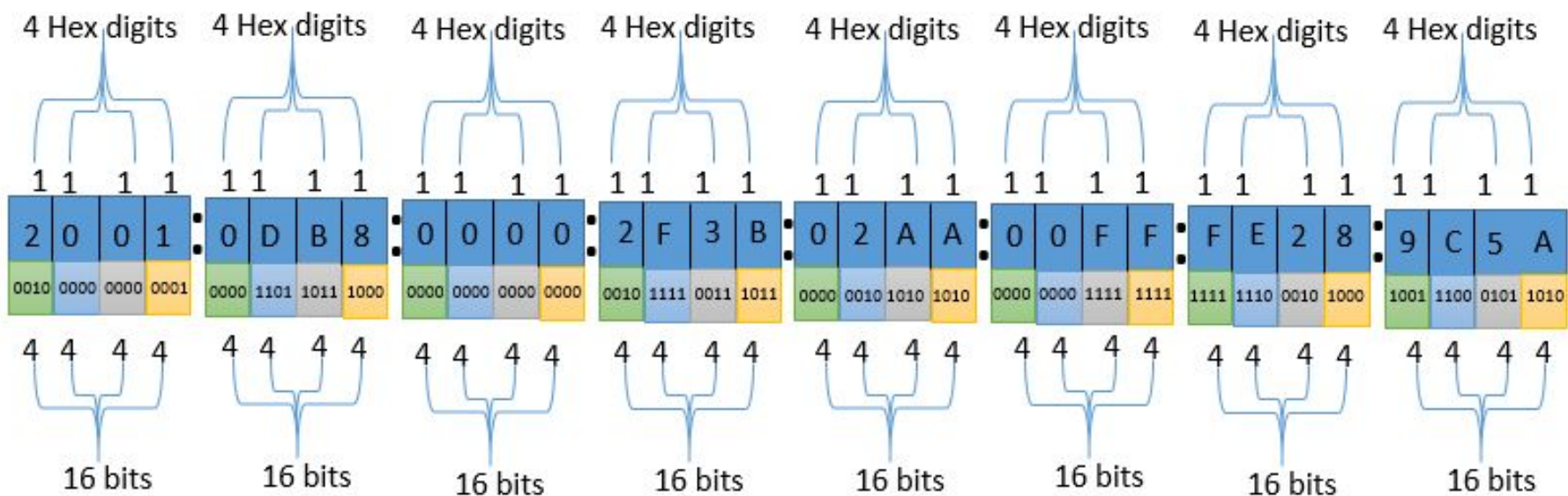
- Called string notation



# IPv6 Address

## IPv6 address in Hexadecimal notation

2001:0DB8:0000:2F3B:02AA:00FF:FE28:9C5A



0010000000000000100001101101110000000000000000000001011110011101100000010101010100000000111111111111110001010001001110001011010

## IPv6 address in binary notation

# IPv6 Addressing

- IPv6 Representation – Rule 1:

- The leading zeros in any 16-bit segment do not have to be written. If any 16-bit segment has fewer than four hexadecimal digits, it is assumed that the missing digits are leading zeros.

2031 : 0000 : 130F : 0000 : 0000 : 09C0 : 876A : 130B

2031 : 0 : 130F : 0 : 0 : 9C0 : 876A : 130B

---

8105 : 0000 : 0000 : 4B10 : 1000 : 0000 : 0000 : 0005

8105 : 0 : 0 : 4B10 : 1000 : 0 : 0 : 5

---

0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000

0 : 0 : 0 : 0 : 0 : 0 : 0 : 0

# IPv6 Addressing

- IPv6 Representation – Rule 2:

- Any single, contiguous string of one or more 16-bit segments consisting of all zeroes can be represented once with a double colon.

|                              |
|------------------------------|
| 1080:0:0:0:8:800:200C:417A = |
| FF01:0:0:0:0:0:0:101 =       |
| 0:0:0:0:0:0:0:1 =            |
| 0:0:0:0:0:0:0:0 =            |

# IPv6 Addressing

- IPv6 Representation – Rule 2:

- *Any single, contiguous string of one or more 16-bit segments consisting of all zeroes can be represented once with a double colon.*

Example: **1843::22::fa**

- Illegal because the length of the two all-zero strings is ambiguous.

1843:0000:0000:0000:0022:0000:0000:00fa

or

1843:0000:0000:0022:0000:0000:0000:00fa



# Representing IPv6 addresses



- IPv4 Address



- IPv6 Address
- No Subnet masks in dotted decimal format in IPv6

# The End